

Atharva Agrawal

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Education

Indian Institute of Information Technology, Dharwad **Aug 2023 – Present**
B.Tech in Data Science & Artificial Intelligence **CPI: 8.85** — Minor in **Cybersecurity**
Relevant Coursework: DSA, Object Oriented Programming, DBMS, Operating Systems, Computer Networks, Software Engineering, Cryptography, DevSecOps.

Experience

OMS - Office of Prahlad Joshi (Union Minister, Govt of India) **Dec 2025 – Present**
Full Stack Developer Intern

- Architected and deployed a production-grade **Office Management System (OMS)** from scratch on the **Zoho platform**, automating grievance handling, visitor tracking, and administrative workflows across **6+ operational modules**.
- Designed and implemented an automated **CI/CD pipeline** for build, testing, and deployment workflows, enabling faster release cycles, reducing deployment overhead, and ensuring reliable production deployments.
- Optimized backend queries, API execution, and frontend rendering pipelines, reducing average application latency from **7–8 seconds to nearly 1 second (~85% performance improvement)** and significantly enhancing user experience.
- Built and integrated **15+ RESTful APIs** with **Node.js, Express.js, and Prisma ORM** along with responsive dashboards using **React**, improving internal workflow efficiency and reducing manual processing effort by **60%**.
- Engineered secure **RBAC-based authentication and authorization** supporting **multiple user roles** and securing access across **10+ protected API endpoints** with verification and document-generation workflows.

PearlThoughts

Jul 2025 – Aug 2025

Frontend Developer Intern (Remote)

- Designed a **modular component architecture** with **15+ reusable UI components**, improving maintainability and accelerating development iterations.
- Integrated **mock REST APIs** and implemented **role-based workflows** enabling rapid feature prototyping and **end-to-end workflow testing** across **3 user roles**.

Projects

Kubernetes Attack Path Analyzer



Kubernetes, Docker, React, Python, FastAPI

- Developed a production-grade **Kubernetes Attack Path Analyzer** leveraging **graph algorithms (BFS, DFS, Dijkstra, and Betweenness Centrality)** to detect **46+ potential attack paths**, privilege escalation loops, and critical infrastructure chokepoints in Kubernetes clusters.
- Engineered a scalable full-stack security analytics platform using **FastAPI, React 18, NetworkX, Docker, and Cytoscape.js**, capable of analyzing **500-node Kubernetes graphs in under 1 second** with optimized in-memory graph processing.
- Built and integrated **RESTful security APIs** for attack-path tracing, blast-radius analysis, privilege escalation detection, and AI-generated remediation reporting, achieving **85%+ automated test coverage** across algorithms and end-to-end workflows.
- Designed an AI-powered kill-chain reporting engine using **Gemini 2.0 Flash** to convert complex graph-security analysis into actionable executive-level threat narratives and remediation recommendations for cloud infrastructure security teams.

AI Scholar – Research Paper Question Answering System



React Native, Express.js, FastAPI, MongoDB, ChromaDB

- Built an accessible mobile learning platform that transforms research papers into a **conversational, voice-enabled AI experience**, supporting **natural language queries** and **audio responses**.
- Engineered a **Retrieval-Augmented Generation (RAG) pipeline** sourcing context from a **vector database** to ensure **accurate, citation-backed answers** with minimal hallucinations.
- Delivered **secure authentication, persistent history, streaming responses, Voice-to-Text/Text-to-Speech, dark mode**, and an **automated PDF ingestion engine** for dynamic knowledge expansion.

Technical Skills

Programming Languages: C, C++, TypeScript, JavaScript

Backend: Node.js, Express.js, FastAPI, REST APIs

Frontend: React, React Native, Tailwind CSS

Databases: MySQL, MongoDB, PostgreSQL, Prisma ORM

Tools: Docker, Jenkins, Kubernetes, AWS, Git, Linux, Firebase

Achievements

- LeetCode:** 1600+ contest rating (Solved 500+ questions)
- HackerRank:** 5-star in C++, SQL Certified
- CodeChef:** 3-Star Max(1627)